OBED ARMANDO CAMPOS

FLUID SYSTEMS ENGINEER | EXPERIMENTAL FLUID MECHANICS | PROJECT LEADERSHIP

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PROFESSIONAL SUMMARY

Mechanical engineer with a Ph.D. and hands-on experience leading the design, fabrication, instrumentation, and validation of fluid systems for pulsatile, oscillatory, and biofluid applications. Combines analytical modeling, CFD, experimental diagnostics, and hardware development to translate technical requirements into reliable test systems. Experienced in project planning, cross-functional coordination, design reviews, troubleshooting, laboratory safety, technical documentation, and mentoring engineering teams.

CORE COMPETENCIES

Fluid Systems: pulsatile/oscillatory internal flow, pressure drop, wall shear stress, flow calibration, leakage diagnosis, test-rig design

Project Execution: requirements definition, work planning, milestone tracking, design reviews, test planning, risk identification, documentation, stakeholder communication

Engineering Tools: ANSYS Fluent/Structural, SolidWorks, Inventor, AutoCAD, Fusion 360, MATLAB, Python, C++, Arduino, Linux

Experimental Methods: MRI flow analysis, particle tracking velocimetry, image processing, instrumentation, uncertainty assessment, prototype fabrication

EDUCATION

University of California San Diego (UCSD) San Diego, CA

Ph.D., Engineering Sciences (Mechanical Engineering), June 2026

C.Phil., Engineering Sciences (Mechanical Engineering), December 2024

M.S., Engineering Sciences (Mechanical Engineering), December 2023

B.S., Mechanical Engineering - Specialization in Controls and Robotics, September 2020

PROJECT MANAGEMENT & LEADERSHIP

- Directed end-to-end design-build-test work for experimental fluid systems, converting research objectives into requirements, concepts, fabrication plans, test procedures, validation criteria, and technical deliverables.
- Coordinated work across faculty, postdoctoral researchers, graduate researchers, undergraduate teams, and technical collaborators; communicated progress, surfaced risks, and drove technical issues to resolution.
- Mentored UCSD Mechanical Engineering capstone teams (2023-2025) through scope definition, design reviews, prototype development, testing, and troubleshooting.
- Served as Area Safety Coordinator for the Bio-Fluidics Laboratory (2023-2025), supporting safe operations, procedure development, equipment readiness, and hazard awareness.

RESEARCH & ENGINEERING EXPERIENCE

Multiscale Flow Physics Lab | Graduate Student Researcher | UCSD Apr 2022-Jun 2026

Advisor: Antonio L. Sánchez, Professor, Mechanical and Aerospace Engineering

- Led the design, manufacture, and validation of a canonical spinal-canal model for MRI-based cerebrospinal-fluid flow measurements.
- Planned experimental campaigns and integrated MRI data, custom MATLAB/Python processing, theoretical models, and computational analysis to characterize oscillatory flow and cycle-to-cycle transport.
- Coordinated experimental setup, calibration, data quality checks, troubleshooting, and publication documentation with multidisciplinary collaborators.

Experimental Bio-Fluidics Lab | Graduate Student Researcher | UCSD Sep 2021-Jun 2026

Advisor: Geno R. Pawlak, Professor, Mechanical and Aerospace Engineering

- Led development and testing of pulsatile flow systems that applied controlled shear stress, pressure, and mechanical loading to endothelial-cell test chambers.
- Used analytical models and ANSYS Fluent to predict flow rate, pressure distribution, and wall shear stress; compared predictions with experiments to validate system performance.
- Owned fabrication, assembly, instrumentation, calibration, test execution, and root-cause troubleshooting for leakage, alignment, flow distribution, and measurement issues.
- Developed technical procedures, analyzed uncertainty, communicated findings in design reviews and conferences, and contributed to peer-reviewed publication.

Lasheras-del Alamo Biomechanics Laboratory | Researcher/Consultant | UCSD Feb 2019-Jun 2021

Advisors: Juan Carlos del Alamo and Juan C. Lasheras, Mechanical and Aerospace Engineering

- Managed multiple prototype-development efforts for biomechanical test systems, balancing mechanical design, electronics, controls, fabrication constraints, biological requirements, and validation needs.
- Consulted with the Pulmonary Hypertension research group on devices that apply combined mechanical loads to pulmonary artery endothelial cells.

SELECTED FLUID-SYSTEM & HARDWARE PROJECTS

Multi-Chamber Pulsating-Flow Device 2021-2026
- Designed and validated a serial multi-chamber system for controlled pulsatile pressure and spatially uniform shear stress across endothelial-cell chambers.
- Integrated theory, CFD, CAD, fabrication, instrumentation, and experiments; documented the system in a peer-reviewed Journal of Biomechanical Engineering publication.

Spinal-Canal Flow Phantom for Time-SLIP MRI 2022-2026
- Designed a canonical annular flow model and experimental protocol to evaluate MRI-based measurements of oscillatory displacement and Lagrangian drift.
- Developed custom image-analysis workflows and assessed resolution, signal quality, processing sensitivity, and measurement uncertainty.

Flow Chamber with Stretching Boundary 2023-2026
- Developed a theoretical and experimental framework for flow over a cyclically stretching membrane as a foundation for combined shear, pressure, and stretch loading.
- Built analytical models, compared predictions with dye-tracking experiments, and diagnosed leakage and geometric sources of flow asymmetry.

Multiplex Cell Stretcher Feb-Jun 2019
- Designed and manufactured a prototype for simultaneous cyclic stretching of four PDMS chambers used for long-duration adherent-cell culture.
- Co-developed standard operating procedures, validated cyclic stretching with HUVECs, and created Arduino-based controls with a user interface.

Enclosed Cell Stretcher Sep 2019-Jun 2020
- Developed a prototype for cyclic stretching under pulsatile flow; designed mechanical components in SolidWorks and generated Tormach CNC toolpaths in Fusion 360.
- Designed Arduino-based electronics allowing users to modify cyclic-stretching parameters.

Microfluidic Endothelial-Cell Test System | Capstone Project Mar-Jun 2020
- Collaborated on a four-chamber fluid system with electronically controlled valves, pressure sensors, servo-driven mechanisms, and a user interface for regulating pressure drops.

Hydrogen-Bubble Flow Visualization System Mar-Jun 2020
- Designed an Arduino-controlled pulsed-DC system for generating hydrogen bubbles between electrodes to visualize flow patterns in a test chamber.

Bioreactor Temperature-Control System Jun 2019-Dec 2020
- Designed an Arduino-based closed-loop electronics system that autonomously regulated live-imaging bioreactor temperature at a defined set point.

PDMS Coating Device Sep 2019-Jan 2020
- Designed electronics and multi-axis motion controls for rotating wax models at adjustable speeds during PDMS coating.

PUBLICATIONS

1. Campos, O., et al. (2026). Phantom-Based Evaluation of Time-SLIP MRI for Measuring Spinal-Canal Lagrangian Drift. Submitted to Scientific Reports.
2. Campos, O. A., Garcia-Herreros, A., Sánchez, A. L., Fineman, J. R., & Pawlak, G. (2024). A Multichamber Pulsating-Flow Device With Optimized Spatial Shear Stress and Pressure for Endothelial Cell Testing. Journal of Biomechanical Engineering, 147. https://doi.org/10.1115/1.4066800
3. Sincomb, S., et al. (2024). Oscillatory Flow in the Cerebral Aqueduct. European Journal of Mechanics - B/Fluids. https://doi.org/10.1016/j.euromechflu.2024.01.010
4. Schwartz, A. B., Campos, O. A., et al. (2021). Elucidating the Biomechanics of Leukocyte Transendothelial Migration by Quantitative Imaging. Frontiers in Cell and Developmental Biology, 9, 635263. https://doi.org/10.3389/fcell.2021.635263

TEACHING & WORK EXPERIENCE

Teaching Assistant | UCSD Spring 2022; Fall 2024; Fall 2025

- Supported instruction, assessment, student feedback, and technical communication for mechanical engineering coursework.

Reader | UCSD Fall 2021; Spring 2022

- Evaluated technical assignments and provided consistent, timely feedback using course grading standards.

Building Maintenance Worker | UCSD Recreation Sep 2016-May 2018

- Performed hands-on facility maintenance and supported safe, reliable operation of recreation buildings and equipment.

LEADERSHIP, SERVICE & TECHNICAL COMMUNICATION

- Executive Board Member, Latine/x Scholar Network | 2024-Present
- Presenter, APS Division of Fluid Dynamics Annual Meeting | 2022, 2023, 2024
- Mentor, GEAR and ISRP Programs at UCSD | 2023-2024
- Member, American Institute of Aeronautics and Astronautics | 2018-2019
- Member, Society of Hispanic Professional Engineers | 2017-2018
- Member, Rocket Propulsion Laboratory New Member Project | 2017-2018

TECHNICAL SKILLS

Fluid Mechanics & Testing: pulsatile and oscillatory flow, internal flows, pressure drop, wall shear stress, calibration, flow visualization, uncertainty analysis

Simulation & CAD: ANSYS Fluent, ANSYS Structural, SolidWorks, Inventor, AutoCAD, Fusion 360

Programming & Controls: MATLAB, Python, C++, Arduino, Linux, embedded control, data acquisition and automation

Data & Imaging: particle tracking velocimetry, MRI flow analysis, image processing, signal analysis, technical plotting

Fabrication: CNC milling, manual mill and lathe, laser cutting, plastic molding, bandsaw, drill press, prototype assembly

Project Management: requirements, schedules and milestones, design reviews, test plans, risk tracking, SOPs, technical reports, mentoring

LANGUAGES

English (native) | Spanish (California State Seal of Biliteracy)